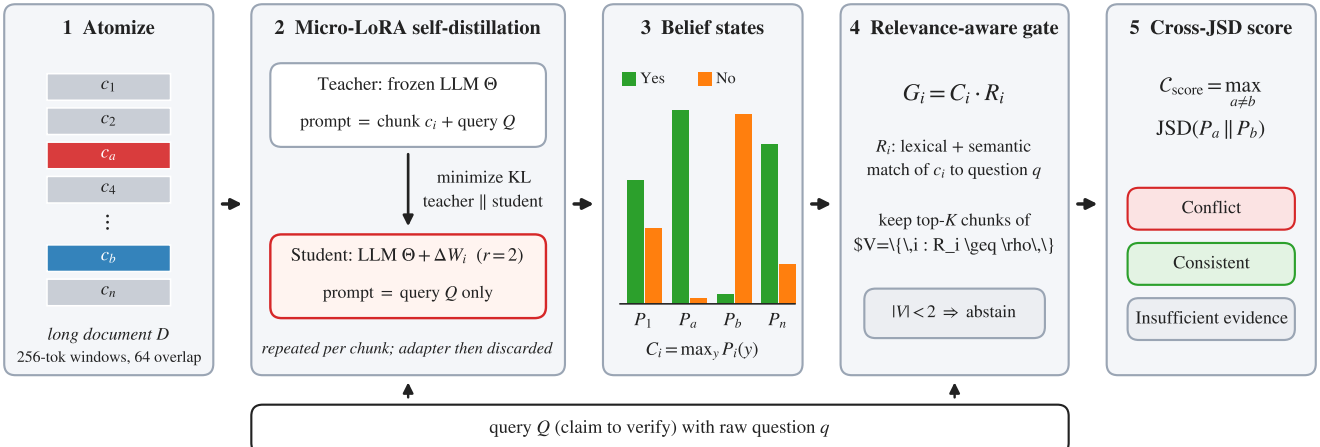

1 Atomize
c_1
c_2
c_a
c_4
c_b
c_n
long document D
256-tok windows, 64 overlap
2 Micro-LoRA self-distillation
Teacher: frozen LLM Θ
prompt = chunk c_i + query Q
minimize KL
teacher ‖ student
Student: LLM Θ + ΔW_i (r = 2)
prompt = query Q only
repeated per chunk; adapter then discarded
3 Belief states
Yes
No
P_1 P_a P_b P_n
$C_i = \max_y P_i(y)$
4 Relevance-aware gate
$G_i = C_i \cdot R_i$
R_i: lexical + semantic match of c_i to question q
keep top-K chunks of
$V=\{\,i : R_i \geq \rho\,\}
|V| < 2 ⇒ abstain
5 Cross-JSD score
$C_{\text{score}} = \max_{a \neq b}$
$\text{JSD}(P_a \,\|\, P_b)$
Conflict
Consistent
Insufficient evidence
query Q (claim to verify) with raw question q